

1 B) Cylinders

- Standard, it never hurts

Displacement

- Standard, R gets too big

HorsePower

- Standard, Ranged numbers.

weight

- // , // //

Acceleration

- // , // //

model year

- Standard ~~name/slot~~ ~~could~~
~~potentially work but~~

origin

- one-hot, non-numerical
Semantics.

C) One-hot between all names in dataset.

D) If we only want one good hypothesis
then we could use test data error.

If we want a good learning algo
for all h we would X-validate

2

C) Reorderings of words produce the same vector.

D) Since ordering doesn't matter and we lose interpretability, we could drop them.

3

A)
$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

C) It is not clear if the data will be linearly separable. But no information is lost. Just harder to interpret.

D) using ~~no~~ ^{black} of pixels in each row.

4A) it might be very hard. ~~Wrong~~

4B) Bucket the frequencies and use
a binary "multi-hot" vector.